Cyclohexane Reactor Ver 1 · operator training simulator

Equipment list

Document no. CHX-V1-ME-EQL-001 · **Discipline** Mechanical engineering

Client Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University (teaching)

• 17 equipment items (2 HOLD) with service, type, size, operating and design conditions, material, weight estimate, motor rating and datasheet number.
• Design pressures from the relief sets of the Safety-layers basis (loop 36 bar g, K-101 discharge 42 bar g, D-101 8.0 bar g, cooling water 10 bar g); design temperatures from the upsets.
• ASME VIII Div. 1 wall thickness for 12 shells; exchanger rating of E-101 (fouling margin), E-103 and E-106; agitator M-101; hot-hydrogen check.
• Datasheets in CHX-V1-ME-DS-001 (15 sheets CHX-V1-ME-DS-). Checks against the model and the other documents: 109 ✓, 9 ✗; 11 engineering queries.

Revisions

Rev	Date	Description	Originator	Checker	Approver
0	2026-10-26	Issued for teaching	Apinan Soottitantawat		

This document is generated for teaching from the simulator model of the unit. It is not issued for construction; every number comes from the model or is labelled assumed / typical practice. No client, contractor or vendor is implied.

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1 · Basis

This equipment list is written from the simulator model of Cyclohexane Reactor Ver 1 at its design point (noise off) and from the other teaching documents: the heat and material balance CHX-V1-PE-HMB-001 and line list CHX-V1-PE-LL-001 (Process engineering), the Design tab sizing (R-101 volume and H/D, E-101 area, R-103 bed), the relief sets of the Safety-layers tab (SL_ASSUMED), the electrical load list CHX-V1-EE-LL-001 and the EPC P&ID CHX-V1-PID-001. Every other number is labelled assumed or typical practice.

Rule	Applied as
Design pressure	≥ the set of the relief device protecting the item: hydrogen loop 36 bar g (PSV-101/102/104); K-101 discharge 42 bar g (PSV-112); D-101 8.0 bar g (PSV-109); pump casings relief set + shut-off rise; cooling-water sides 10 bar g
Design temperature	maximum operating + 30 °C rounded up (typical), or the upset temperature (loss of cooling, trip setting, boiling ceiling of the reactor); R-101/R-103 from SL_ASSUMED (250/260 °C)
Code	ASME VIII Div. 1: UG-27 cylinder t = PR/(SE − 0.6P) + CA; UG-32 2:1 ellipsoidal head t = PD/(2SE − 0.2P) + CA; hydrotest UG-99(b) 1.3 × P × S(test)/S(design)
Materials	killed CS SA-516 Gr 70 N with PWHT for hot hydrogen and Raney-Ni caustic residue; 316L where no corrosion allowance can be taken (E-101 tubes, D-102, candles, agitator); CA 3 mm process, 1.5 mm water
Flanges	ASME B16.5 group 1.1: class from the design pressure at the design temperature (typical table)
Nozzles	sized to the line list CHX-V1-PE-LL-001; EPC P&ID line shown beside it (differences: EQ-em-4)
Weights	steel from the chosen plates × allowance for nozzles, internals and supports; machines order of magnitude (assumed)

2 · Equipment list

Tag	Item / service	Type	Size / capacity	Operating	Design	Material · CA	Weight t	Motor kW	Datasheet
R-101	Main hydrogenation reactor (slurry CSTR) Benzene + H_2 → cyclohexane on Raney-Ni slurry; pump-around cooling through E-101, boil-up to E-103	Vertical agitated pressure vessel, 2:1 ellipsoidal heads, skirt	ID 3,800 × T/T 4,500 mm 65.4 m³ total; 29.0 m³ liquid at NLL (LI-101 65 %)	30.00 bar g · 200.0 °C	36.0 bar g · 250 °C	SA-516 Gr 70 N + PWHT (killed CS) CA 3 mm	49.0 / 66.6	M-101: 250 (6.6 kV)	DS-R-101
M-101	R-101 agitator Suspends the Raney-Ni, disperses the hydrogen from the sparger	Top-entry agitator: gas-dispersing turbine + upper axial impeller, gearbox, VSD	Impellers Ø1.52 / Ø1.71 m at 128 rpm 180 kW ungassed, 153 kW gassed (model)	—	—	Wetted parts 316L (shaft, impellers); double mechanical seal with barrier fluid (typical)	6.0 / 6.0	M-101: 250 (6.6 kV)	DS-R-101
R-102	Finishing slurry reactor (jacketed) Converts the remaining benzene (≈2 % → ≈100 ppm); jacket on MP steam (HEAT) or CW (COOL) by HS-102	Vertical agitated pressure vessel with jacket, 2:1 heads, skirt	ID 3,200 × T/T 3,800 mm 39.1 m³ total; 15.0 m³ liquid at NLL (LI-102 55 %)	29.60 bar g · 160.0 °C	36.0 bar g · 250 °C	SA-516 Gr 70 N + PWHT (killed CS); jacket SA-516 Gr 70 CA 3 mm	31.8 / 41.4	M-102: HOLD	DS-R-102
M-102	R-102 agitator (HOLD) Keeps the Raney-Ni suspended in R-102 and disperses the V-103 gas Kept on the list as drawn on the P&ID (sheet 2) — HOLD until Process states how R-102 keeps the catalyst suspended (EQ-em-7).	Top-entry agitator (as drawn on the P&ID); not in the model nor the load list	HOLD ≈ 23 kW at 1.5 kW/m³ (typical, assumed) → 30 kW motor	—	—	Wetted parts 316L (typical)	2.0 / 2.0	M-102: HOLD	DS-R-102

Tag	Item / service	Type	Size / capacity	Operating	Design	Material · CA	Weight t	Motor kW	Datasheet
R-103	Trickle-bed finishing reactor Adiabatic polishing of benzene to ≈2 ppm on supported Ni; liquid from F-102/LV-102 and gas from R-102 in co-current down-flow	Vertical fixed-bed vessel, 2:1 heads, skirt; distributor tray, support grid, inert balls	ID 1,950 × T/T 7,400 mm; bed 5.71 m 16.2 m³ catalyst (LHSV 3 h^{-1}, Design tab)	29.25 bar g · 158.6 °C	36.0 bar g · 260 °C	SA-516 Gr 70 N + PWHT (killed CS); internals 304L CA 3 mm	17.0 / 34.6	—	DS-R-103
E-101	Pump-around cooler / LP steam generator Tubes: R-101 slurry 200 → 159.3 °C; shell: boiling BFW at the D-101 pressure (COOL) or MP steam (HEAT, start-up)	Shell and tube, TEMA AEL, horizontal; 4 tube passes; natural circulation (thermosyphon) to D-101	Shell ID 1,250 mm × 1210 tubes Ø25.4 × 7 320 mm 9.60 MW (model); A 707 m²	32.00 bar g · 200.0 °C other side 4.50 bar g	40.0 bar g · 250 °C other side 12.0 · 200 °C	Tubes SA-213 TP316L (slurry erosion, no CA); channels SA-516 Gr 70 + PWHT CA 3; shell SA-516 Gr 70 CA 3 CA 3 mm	19.3 / 26.1	—	DS-E-101
E-103	Reflux condenser (cooling water) Condenses the cyclohexane boiled up in R-101 out of the hydrogen; condensate refluxes via V-103	Shell and tube, TEMA AES, horizontal; CW in 2-pass tubes, process vapour on the shell	Shell ID 950 mm × 991 tubes Ø19.05 × 6 100 mm 7.87 MW (HMB); A 362 m²	30.00 bar g · 200.0 °C other side 4.00 bar g	36.0 bar g · 250 °C other side 10.0 · 65 °C	Shell SA-516 Gr 70 N + PWHT CA 3; tubes SA-214 (CS) CA 0 with CW treatment (typical); channels CS CA 1.5 CA 3 mm	8.9 / 11.6	—	DS-E-103
E-106	Product cooler (cooling water) Cools the R-103 effluent (gas + liquid) to the V-104 temperature	Shell and tube, TEMA AES, horizontal; CW in 6-pass tubes	Shell ID 950 mm × 936 tubes Ø19.05 × 6 100 mm 3.49 MW (HMB, sensible estimate); A 342 m²	29.25 bar g · 158.6 °C other side 4.00 bar g	36.0 bar g · 260 °C other side 10.0 · 65 °C	Shell SA-516 Gr 70 N + PWHT CA 3; tubes SA-214 (CS); channels CS CA 1.5 CA 3 mm	8.6 / 11.2	—	DS-E-106
D-101	Steam drum Raises LP steam from E-101 by natural circulation; PIC-110 / PV-110 to the LP header	Horizontal drum, 2:1 heads, saddles; steam separators and demister	ID 2,400 × T/T 8,400 mm water span 24 t (model K.Mspan) = 24.9 m³ between LT-109 0 and 100 %	4.50 bar g · 155.8 °C	8.0 bar g · 180 °C	SA-516 Gr 70 CA 1.5 CA 1.5 mm	8.6 / 26.1	—	DS-D-101
D-102	Catalyst charge pot Holds a Raney-Ni charge under liquid and N_2; charged to R-101 through HV-117 Charging method and pressure source to confirm (EQ-em-8).	Vertical pot, 2:1 heads, conical-bottom option, legs (assumed size)	ID 1,000 × T/T 1,500 mm (assumed) 1.44 m³ (one make-up batch, assumed)	0.00 bar g · 40.0 °C	36.0 bar g · 250 °C	SA-240 316L (pyrophoric Ni, caustic residue; no CA) CA 0 mm	1.5 / 3.8	—	DS-D-102
V-103	Reflux drum Separates the E-103 condensate (reflux to R-101 by gravity) from the hydrogen to R-102	Vertical drum, 2:1 heads, mesh demister, skirt	ID 1,200 × T/T 3,000 mm (PFD size) 3.8 m³; gas 0.20 m/s vs 0.68 allowed (120 %)	30.00 bar g · 55.7 °C	36.0 bar g · 250 °C	SA-516 Gr 70 N + PWHT (killed CS) CA 3 mm	2.9 / 4.3	—	DS-V-103
V-104	HP separator Separates the product liquid from the recycle hydrogen at 29 bar g; make-up H_2 enters here (PFD)	Horizontal drum, 2:1 heads, saddles; inlet device, mesh demister, vortex breaker	ID 2,300 × T/T 9,100 mm level span 31.0 m³ (model K.V4span 30 m³)	29.00 bar g · 44.9 °C	36.0 bar g · 190 °C	SA-516 Gr 70 N (killed CS) CA 3 mm	27.3 / 39.6	—	DS-V-104
F-102A/B	Catalyst candle filters (1 duty + 1 standby) Hold back the Raney-Ni from the R-102 liquid before R-103; PDI-102	Vertical filter housing, bolted top cover, sintered-metal candles, back-flush (assumed)	ID 800 × T/T 2,400 mm; 48 candles Ø60 × 1500 mm (assumed) 13.6 m²; 2.41 m³/m²·h at design	32.60 bar g · 160.0 °C	40.0 bar g · 250 °C	SA-516 Gr 70 N + PWHT CA 3; candles sintered 316L CA 3 mm	1.6 / 2.1	—	DS-F-102

Tag	Item / service	Type	Size / capacity	Operating	Design	Material · CA	Weight t	Motor kW	Datasheet
P-100A/B	Benzene feed pumps TK-100 → FIC-101/XV-101 → R-101	Centrifugal, multistage between-bearings (API 610 BB3/BB4 type, typical)	rated 48.7 m³/h at 374 m ΔP 31.4 bar; absorbed ≈ 71 kW (η 0.60 assumed)	30.00 bar g · 40.0 °C other side 1.00 bar g	40.0 bar g · 70 °C	Casing CS (S-6 class, API 610 typical); dual mechanical seals (plan 53B, assumed) CA 3 mm	3.5 / 3.5	P-100A/B: 110	DS-P-100
P-101A/B	Pump-around pumps R-101 → FV-104 → E-101 / by-pass → R-101 (always K.Fpa)	Centrifugal, overhung centreline-mounted (API 610 OH2 type, typical), slurry service	rated 998 m³/h at 33.6 m ΔP 2.0 bar; absorbed ≈ 79 kW (η 0.70 assumed); NPSHa ≈ 3.9 m (Process engineering)	32.00 bar g · 200.0 °C other side 30.00 bar g	40.0 bar g · 250 °C	Casing CS + hard-faced wear parts (slurry); dual pressurised seals (plan 53B/54, typical for hot flashing slurry) CA 3 mm	5.0 / 5.0	P-101A/B: 200	DS-P-101
P-102A/B	R-102 transfer pumps R-102 → F-102A/B → LV-102 → R-103	Centrifugal, overhung (API 610 OH2 type, typical), slurry service	rated 39.5 m³/h at 48 m ΔP 3.0 bar (K.pumpDP); absorbed ≈ 6.6 kW (η 0.50 assumed)	32.60 bar g · 160.0 °C other side 29.60 bar g	40.0 bar g · 250 °C	Casing CS + hard-faced wear parts; dual seals (typical) CA 3 mm	1.2 / 1.2	P-102A/B: 37	DS-P-102
K-101	Recycle hydrogen compressor V-104 gas → FIC-103/FV-103 → R-101 sparger	Centrifugal, 1–2 impellers, dry gas seals (typical for low-ratio recycle H_2; type to be confirmed, EQ-ee-1)	46,000 Nm³/h (120 %: 55,200); ratio 1.100 head 81.6 kJ/kg; absorbed ≈ 194 kW (η 0.75, k 1.4 assumed) vs model econ() 1,200 kW	32.00 bar g · 60.0 °C other side 29.00 bar g	42.0 bar g · 90 °C	Casing CS (killed); impellers 17-4PH (typical) CA 3 mm	35.0 / 35.0	K-101: 1600 (6.6 kV)	DS-K-101

Weight: empty / operating, estimates. Motor ratings from CHX-V1-EE-LL-001. Datasheet numbers CHX-V1-ME-DS-<tag> in CHX-V1-ME-DS-001.

3 · Design conditions and their sources

Tag	Max. operating / upset	Design P	Source of the design pressure	Protecting device	Design T	Source of the design temperature
R-101	30.44 bar g; 258.3 °C — TSHH-100 trip 220 °C; boiling ceiling TbC(PI-101) 258.3 °C	36.0 bar g	SL_ASSUMED.Pdes = PSV-101 set	PSV-101 36.0 bar g	250 °C	SL_ASSUMED.Tdes1 (critical limit of the R-101 ladder)
R-102	29.86 bar g; 184.0 °C — jacket MP steam 184 °C (K.Tms)	36.0 bar g	SL_ASSUMED.Pdes = PSV-102 set	PSV-102 36.0 bar g	250 °C	as R-101 (SL_ASSUMED.Tdes1); covers MP steam 184 °C
R-103	29.36 bar g; 220.0 °C — TSHH-107 trip 220 °C	36.0 bar g	SL_ASSUMED.Pdes = PSV-102 set (no isolation from R-102)	PSV-102 36.0 bar g	260 °C	SL_ASSUMED.Tdes3 (critical limit of the bed ladder)
E-101	32.44 bar g; 220.0 °C — slurry from R-101; shell up to MP steam 184 °C (HEAT)	40.0 bar g	tubes: R-101 PSV 36 + P-101 shut-off rise 4 bar (as line 16"-P-1005-B3); shell: MP steam line 12 bar g (8"-S-3002-A1)	PSV-101 + P-101 shut-off 40.0 bar g	250 °C	tubes SL_ASSUMED.Tdes1; shell = MP steam line design 200 °C
E-103	30.44 bar g; 220.0 °C — R-101 overhead; on loss of CW the drum side sees 200 °C gas	36.0 bar g	shell SL_ASSUMED.Pdes (PSV-101, no isolation); tubes CW 10 bar g (TSV set)	PSV-101 36.0 bar g	250 °C	shell SL_ASSUMED.Tdes1; tubes CW supply line 65 °C (LL)
E-106	29.36 bar g; 220.0 °C — R-103 outlet; TSHH-107 220 °C	36.0 bar g	shell SL_ASSUMED.Pdes (PSV-104); tubes CW 10 bar g	PSV-104 36.0 bar g	260 °C	shell = R-103 outlet line (SL_ASSUMED.Tdes3); tubes 65 °C
D-101	4.50 bar g; 175.7 °C — saturation at the PSV-109 set 175.7 °C	8.0 bar g	SL_ASSUMED.Psv109 = PSV-109 set	PSV-109 8.0 bar g	180 °C	saturation at the set + margin (as 10"-S-3001-A1)
D-102	30.00 bar g; 200.0 °C — sees R-101 when HV-117 is open	36.0 bar g	as R-101 (connected through HV-117)	PSV-101 (when HV-117 is open) 36.0 bar g	250 °C	as R-101
V-103	30.44 bar g; 200.0 °C — loss of CW to E-103: R-101 overhead gas at 200 °C	36.0 bar g	SL_ASSUMED.Pdes (PSV-101, no isolation)	PSV-101 36.0 bar g	250 °C	as the E-103 outlet line (loss of cooling)
V-104	29.00 bar g; 158.6 °C — loss of E-106 cooling: R-103 outlet 158.6 °C (EQ-ep-12)	36.0 bar g	SL_ASSUMED.Pdes = PSV-104 set	PSV-104 36.0 bar g	190 °C	R-103 outlet + 30 °C (loss of E-106 cooling, typical rule)
F-102A/B	32.86 bar g; 184.0 °C — as R-102	40.0 bar g	P-102 discharge side (as 3"-P-1011-B3; EQ-ep-5 resolution)	TSV-102F 36.0 bar g	250 °C	as R-102
P-100A/B	35.99 bar g; 40.0 °C — pressure: shut-off (model: no feed above 35 bar g); temperature: benzene feed	40.0 bar g	casing = discharge line 3"-P-1002-B1 (40 bar g)	P-100 shut-off 36.0 bar g	70 °C	as the line
P-101A/B	40.00 bar g; 220.0 °C — pressure: R-101 PSV set + shut-off rise 4 bar; temperature: TSHH-100 trip	40.0 bar g	as 16"-P-1004-B3	PSV-101 + P-101 shut-off 40.0 bar g	250 °C	SL_ASSUMED.Tdes1
P-102A/B	39.00 bar g; 184.0 °C — pressure: R-102 PSV set + K.pumpDP; temperature: jacket MP steam	40.0 bar g	as 3"-P-1011-B3	PSV-102 + P-102 rise (K.pumpDP) 39.0 bar g	250 °C	SL_ASSUMED.Tdes1
K-101	32.82 bar g; 60.0 °C — pressure: highest discharge (70 % case); temperature: gas to the sparger (model)	42.0 bar g	SL_ASSUMED.PdesK = PSV-112 set (deadhead from 35 bar g suction × (1 + K.kk))	PSV-112 42.0 bar g	90 °C	as 8"-H-2008-B2

Boiling ceiling: R-101 at PI-101 boils at TbC = 258.3 °C (model), 272.4 °C at the PSV set — above the R-101 design temperature 250 °C (EQ-em-1).

Hot hydrogen (schematic Nelson-type check)

Item	T °C (trip)	pH_2 bar a	CS limit °C (schematic)	Margin K	Result
R-101 (liquid in equilibrium with the gas)	200 (220)	13.8	289	69	✓
R-101 overhead / E-103 inlet	200 (220)	13.8	289	69	✓
R-102	160	18.6	278	118	✓

Item	T °C (trip)	pH_2 bar a	CS limit °C (schematic)	Margin K	Result
R-103 bed	159 (220)	18.6	278	58	✓
E-106 inlet	159	18.6	278	119	✓
V-104 / K-101 suction	45	27.6	266	221	✓
K-101 discharge / sparger	60	30.4	264	204	✓

Curve: schematic shape after API 941, approximate, for teaching only (assumed); practice keeps ≥ 28 °C below the curve. Result: killed CS with PWHT for the hot-hydrogen items (answers EQ-ep-11).

4 · Wall thickness (ASME VIII Div. 1)

Worked example R-101: P = 36 bar g = 3.6 MPa, T = 250 °C, ID 3,800 mm, SA-516 Gr 70 S = 138 MPa, E = 1, CA 3 mm. Shell t = 3.6 × 1,903 / (138 × 1 − 0.6 × 3.6) + 3 = **53.4 mm** → plate **55 mm**. Head t = 3.6 × 3,806 / (2 × 138 − 0.2 × 3.6) + 3 = **52.8 mm**, + 10 % forming → **60 mm**. MAWP 36.9 bar g; hydrotest 46.8 bar g. P ≤ 0.385·S·E and t ≤ R/2: the thin-wall formula applies.

Item	P bar g	T °C	ID mm	Material	S MPa	E	CA mm	Shell req. / plate mm	Head req. / plate mm	MAWP bar g	Hydrotest bar g	OK
R-101	36.0	250	3,800	SA-516-70	138	1	3	53.4 / 55	52.8 / 60	36.9	46.8	✓
R-102	36.0	250	3,200	SA-516-70	138	1	3	45.5 / 50	44.9 / 50	36.2	46.8	✓
R-103	36.0	260	1,950	SA-516-70	138	1	3	28.9 / 30	28.6 / 32	36.3	46.8	✓
V-103	36.0	250	1,200	SA-516-70	138	1	3	19.0 / 20	18.8 / 22	37.6	46.8	✓
V-104	36.0	190	2,300	SA-516-70	138	1	3	33.6 / 36	33.2 / 40	38.5	46.8	✓
D-101	8.0	180	2,400	SA-516-70	138	0.85	1.5	9.7 / 12	9.7 / 12	8.9	10.4	✓
D-102	36.0	250	1,000	SA-240-316L	103	1	0	17.9 / 20	17.5 / 20	36.8	52.3	✓
F-102A/B	40.0	250	800	SA-516-70	138	1	3	14.9 / 16	14.7 / 18	42.7	52.0	✓
E-101 shell	12.0	200	1,250	SA-516-70	138	0.85	3	9.5 / 10	9.4 / 10	12.4	15.6	✓
E-101 channel	40.0	250	1,250	SA-516-70	138	1	3	21.5 / 22	21.3 / 22	40.4	52.0	✓
E-103 shell	36.0	250	950	SA-516-70	138	1	3	15.7 / 16	15.5 / 16	36.1	46.8	✓
E-106 shell	36.0	260	950	SA-516-70	138	1	3	15.7 / 16	15.5 / 16	36.1	46.8	✓

Allowable stresses, joint efficiencies and the plate list are typical teaching values (assumed). External pressure, nozzle reinforcement, wind, seismic and the supports are not shown.

5 · Exchanger rating

E-101 pump-around cooler / LP steam generator

Model relation: $UA = UA_1 \cdot (F/1.5)^{0.3}$, boiling shell at T_{sat} ⇒ $\varepsilon = 1 - \exp(-UA/C)$. At design: F = 357 t/h, UA 600 kW/K, NTU 2.54, ε 0.921, Q = 9,598 kW, LMTD 16.0 K. Area 706 m² (Design tab, U 0.85 kW/m²K assumed) → 1210 tubes Ø25.4 × 2.11 × 7 320 mm 316L, 4 passes, shell 1,250 mm; tube velocity 1.49 m/s (full flow 2.53, 70 % case 0.94); tube-side ΔP 0.33 bar (allowable 0.64 bar, Process engineering).

Design margin. Clean capacity at the full P-101 flow 14,619 kW = 1.52 × design. UA may fall to **45 %** of clean (fouling resistance 1.25 m²K/kW) before the design duty needs the whole P-101 flow; 62 % at the 120 % case. A typical design fouling allowance for this service, 0.35 m²K/kW (assumed), is well inside the margin.

E-103 and E-106

Item	Duty kW	Process °C	CW °C	LMTD K	F	U kW/m²K	A req./inst. m²	Capacity kW	120 % duty / capacity kW	Tubes, passes, shell	CW v m/s
E-103	7,872	200.0 → 55.7	32 → 47.7	69.1	0.896	0.45	282 / 362	10,086	9,358 / 10,296	991 × Ø19.05 × 6.1 m, 2 p, 950 mm	1.25
E-106	3,492	158.6 → 44.9	32 → 45.9	46.0	0.808	0.4	235 / 342	5,085	4,192 / 4,615	936 × Ø19.05 × 6.1 m, 6 p, 950 mm	1.98

U assumed (typical for hydrogen-rich partial condensation / cooling); installed area = larger of 1.2 × design and 1.1 × the 120 % case. The model condenser E-103 has UA_3 500 kW/K vs 163 kW/K here (EQ-em-9).

6 · Agitator M-101

$P = N_p \cdot \rho \cdot N^3 \cdot D^5$. Concave-blade gas-dispersing turbine Ø1.52 m (D/T 0.40, N_p 3.2) + hydrofoil Ø1.71 m (N_p 0.3): N_p total 3.74, slurry ρ 607 kg/m³ → 128 rpm for the model ungassed power 180 kW; tip speed 10.2 m/s; gassed 153 kW (JI-101), ratio 0.85 (0.7–0.9 typical for a concave-blade turbine; 0.4–0.6 for a flat-blade disc turbine). Gas 0.723 m³/s at the sparger: Fl 0.096 < flooding 2.15. 6.2 kW/m³ ungassed. Motor 250 kW VSD (load list) ≥ model maximum 204.3 kW. M-102 (R-102) HOLD: ≈ 23 kW → 30 kW (EQ-em-7).

7 · Datasheet register

Datasheet	Title	EQL item	Nozzles	Package
CHX-V1-ME-DS-R-101	Main hydrogenation reactor with agitator M-101	R-101	17	PK-01
CHX-V1-ME-DS-R-102	Finishing slurry reactor (jacketed) with agitator M-102 (HOLD)	R-102	16	PK-01
CHX-V1-ME-DS-R-103	Trickle-bed finishing reactor	R-103	8	PK-02
CHX-V1-ME-DS-E-101	Pump-around cooler / LP steam generator	E-101	10	PK-03
CHX-V1-ME-DS-E-103	Reflux condenser (cooling water)	E-103	7	PK-03
CHX-V1-ME-DS-E-106	Product cooler (cooling water)	E-106	7	PK-03
CHX-V1-ME-DS-D-101	Steam drum	D-101	12	PK-04
CHX-V1-ME-DS-D-102	Catalyst charge pot	D-102	7	PK-01
CHX-V1-ME-DS-V-103	Reflux drum	V-103	7	PK-05
CHX-V1-ME-DS-V-104	HP separator	V-104	11	PK-05
CHX-V1-ME-DS-F-102	Catalyst candle filters F-102A/B	F-102A/B	6	PK-08
CHX-V1-ME-DS-P-100	Benzene feed pumps P-100A/B	P-100A/B	2	PK-06
CHX-V1-ME-DS-P-101	Pump-around pumps P-101A/B	P-101A/B	2	PK-06
CHX-V1-ME-DS-P-102	R-102 transfer pumps P-102A/B	P-102A/B	2	PK-06
CHX-V1-ME-DS-K-101	Recycle hydrogen compressor	K-101	2	PK-07

8 · Checks against the model and the other documents

Computed by the same function as the Mechanical tab (emChecks) from the model and the documents: 109 ✓, 9 ✗. A ✗ is a finding recorded as an engineering query.

Check	Item	Value	Limit / reference	OK	Query
Design pressure ≥ relief set	R-101	36.0 bar g	PSV-101 36.0 bar g	✓	
	R-102	36.0 bar g	PSV-102 36.0 bar g	✓	
	R-102 jacket	12.0 bar g	TSV-102J 10.0 bar g	✓	
	R-103	36.0 bar g	PSV-102 36.0 bar g	✓	
	E-101	40.0 bar g	PSV-101 + P-101 shut-off 40.0 bar g	✓	
	E-101 (other side)	12.0 bar g	PSV-109 (COOL only) 8.0 bar g	✓	
	E-103	36.0 bar g	PSV-101 36.0 bar g	✓	
	E-103 (other side)	10.0 bar g	TSV-113 10.0 bar g	✓	
	E-106	36.0 bar g	PSV-104 36.0 bar g	✓	
	E-106 (other side)	10.0 bar g	TSV-106 10.0 bar g	✓	
	D-101	8.0 bar g	PSV-109 8.0 bar g	✓	
	D-102	36.0 bar g	PSV-101 (when HV-117 is open) 36.0 bar g	✓	
	V-103	36.0 bar g	PSV-101 36.0 bar g	✓	
	V-104	36.0 bar g	PSV-104 36.0 bar g	✓	
	F-102A/B	40.0 bar g	TSV-102F 36.0 bar g	✓	
	P-100A/B	40.0 bar g	P-100 shut-off 36.0 bar g	✓	
	P-101A/B	40.0 bar g	PSV-101 + P-101 shut-off 40.0 bar g	✓	
	P-102A/B	40.0 bar g	PSV-102 + P-102 rise (K.pumpDP) 39.0 bar g	✓	
	K-101	42.0 bar g	PSV-112 42.0 bar g	✓	
Design pressure ≥ maximum operating	R-101	36.0 bar g	30.44 bar g (HMB / shut-off)	✓	
	R-102	36.0 bar g	29.86 bar g (HMB / shut-off)	✓	
	R-103	36.0 bar g	29.36 bar g (HMB / shut-off)	✓	
	E-101	40.0 bar g	32.44 bar g (HMB / shut-off)	✓	
	E-103	36.0 bar g	30.44 bar g (HMB / shut-off)	✓	
	E-106	36.0 bar g	29.36 bar g (HMB / shut-off)	✓	
	D-101	8.0 bar g	4.50 bar g (HMB / shut-off)	✓	
	D-102	36.0 bar g	30.00 bar g (HMB / shut-off)	✓	
	V-103	36.0 bar g	30.44 bar g (HMB / shut-off)	✓	
	V-104	36.0 bar g	29.00 bar g (HMB / shut-off)	✓	
	F-102A/B	40.0 bar g	32.86 bar g (HMB / shut-off)	✓	
	P-100A/B	40.0 bar g	35.99 bar g (HMB / shut-off)	✓	
	P-101A/B	40.0 bar g	40.00 bar g (HMB / shut-off)	✓	

Check	Item	Value	Limit / reference	OK	Query
	P-102A/B	40.0 bar g	39.00 bar g (HMB / shut-off)	✓	
	K-101	42.0 bar g	32.82 bar g (HMB / shut-off)	✓	
Design temperature ≥ maximum incl. upsets	R-101	250 °C	258.3 °C · boiling ceiling TbC(PI-101) 258.3 °C; trip 220 °C	✗	EQ-em-1
	R-102	250 °C	184.0 °C · jacket MP steam 184 °C (K.Tms)	✓	
	R-103	260 °C	256.4 °C · boiling ceiling TbC(R-103) 256.4 °C; trip 220 °C	✓	
	E-101	250 °C	220.0 °C · slurry from R-101; shell up to MP steam 184 °C (HEAT)	✓	
	E-103	250 °C	220.0 °C · R-101 overhead; on loss of CW the drum side sees 200 °C gas	✓	
	E-106	260 °C	220.0 °C · R-103 outlet; TSHH-107 220 °C	✓	
	D-101	180 °C	175.7 °C · saturation at the PSV-109 set 175.7 °C	✓	
	D-102	250 °C	200.0 °C · sees R-101 when HV-117 is open	✓	
	V-103	250 °C	200.0 °C · loss of CW to E-103: R-101 overhead gas at 200 °C	✓	
	V-104	190 °C	158.6 °C · loss of E-106 cooling: R-103 outlet 158.6 °C (EQ-ep-12)	✓	
	F-102A/B	250 °C	184.0 °C · as R-102	✓	
	P-100A/B	70 °C	40.0 °C · pressure: shut-off (model: no feed above 35 bar g); temperature: benzene feed	✓	
	P-101A/B	250 °C	220.0 °C · pressure: R-101 PSV set + shut-off rise 4 bar; temperature: TSHH-100 trip	✓	
	P-102A/B	250 °C	184.0 °C · pressure: R-102 PSV set + K.pumpDP; temperature: jacket MP steam	✓	
	K-101	90 °C	60.0 °C · pressure: highest discharge (70 % case); temperature: gas to the sparger (model)	✓	
Wall thickness (ASME VIII-1)	R-101	shell 55 / heads 60 mm	required 53.4 / 52.8 mm at 36 bar g, 250 °C	✓	
	R-102	shell 50 / heads 50 mm	required 45.5 / 44.9 mm at 36 bar g, 250 °C	✓	
	R-103	shell 30 / heads 32 mm	required 28.9 / 28.6 mm at 36 bar g, 260 °C	✓	
	V-103	shell 20 / heads 22 mm	required 19.0 / 18.8 mm at 36 bar g, 250 °C	✓	
	V-104	shell 36 / heads 40 mm	required 33.6 / 33.2 mm at 36 bar g, 190 °C	✓	
	D-101	shell 12 / heads 12 mm	required 9.7 / 9.7 mm at 8 bar g, 180 °C	✓	
	D-102	shell 20 / heads 20 mm	required 17.9 / 17.5 mm at 36 bar g, 250 °C	✓	
	F-102A/B	shell 16 / heads 18 mm	required 14.9 / 14.7 mm at 40 bar g, 250 °C	✓	
	E-101 shell	shell 10 / heads 10 mm	required 9.5 / 9.4 mm at 12 bar g, 200 °C	✓	
	E-101 channel	shell 22 / heads 22 mm	required 21.5 / 21.3 mm at 40 bar g, 250 °C	✓	
	E-103 shell	shell 16 / heads 16 mm	required 15.7 / 15.5 mm at 36 bar g, 250 °C	✓	
	E-106 shell	shell 16 / heads 16 mm	required 15.7 / 15.5 mm at 36 bar g, 260 °C	✓	
Nozzle flange rating (B16.5 group 1.1)	all nozzles	116 of 116	class rating at T_des ≥ P_des	✓	
Nozzle size = line-list size	R-101	7 of 7	CHX-V1-PE-LL-001	✓	

Check	Item	Value	Limit / reference	OK	Query
	R-102	6 of 6	CHX-V1-PE-LL-001	✓	
	R-103	3 of 3	CHX-V1-PE-LL-001	✓	
	E-101	5 of 5	CHX-V1-PE-LL-001	✓	
	E-103	4 of 4	CHX-V1-PE-LL-001	✓	
	E-106	4 of 4	CHX-V1-PE-LL-001	✓	
	D-101	4 of 4	CHX-V1-PE-LL-001	✓	
	V-103	3 of 3	CHX-V1-PE-LL-001	✓	
	V-104	5 of 5	CHX-V1-PE-LL-001	✓	
	F-102	2 of 2	CHX-V1-PE-LL-001	✓	
	P-100	2 of 2	CHX-V1-PE-LL-001	✓	
	P-101	2 of 2	CHX-V1-PE-LL-001	✓	
	P-102	2 of 2	CHX-V1-PE-LL-001	✓	
	K-101	2 of 2	CHX-V1-PE-LL-001	✓	
Nozzle size = EPC P&ID line size	connections on both	25 of 51	CHX-V1-PID-001	✗	EQ-em-4
Every process nozzle has a line-list line	P&ID lines	11 without	CHX-V1-PE-LL-001	✗	EQ-em-5
Hold-up = model	R-101 liquid at NLL (GA)	28.99 m³ clear (32.47 aerated)	model 29.02 m³ (n_1/ρ), ±3 %	✓	
	R-101 LI-101 span (GA)	50.1 m³ to 4.73 m	K.V1span 50 m³ ±3 %, below the TTL 5.45 m	✓	
	R-102 liquid at NLL	15.01 m³ clear	model 15.00 m³ ±3 %	✓	
	V-104 level span	31.0 m³	K.V4span 30 m³, 0 to +10 %	✓	
	D-101 water span	24.6 t	K.Mspan 24 t, 0 to +10 %	✓	
PFD drawing size = EQL	R-101	EQL 3.80 × 4.50 m	PFD 3.6 × 4.5 m ±10 %	✓	
	V-103	EQL 1.20 × 3.00 m	PFD 1.2 × 3 m ±10 %	✓	
	D-101	EQL 2.40 × 8.40 m	PFD 1.4 × 4 m ±10 %	✗	EQ-em-6
	R-102	EQL 3.20 × 3.80 m	PFD 3 × 3.8 m ±10 %	✓	
	R-103	EQL 1.95 × 5.71 m	PFD 1.9 × 5.7 m ±10 %	✓	
	V-104	EQL 2.30 × 9.10 m	PFD 2 × 5 m ±10 %	✗	EQ-em-6
Exchanger rating	E-101 datasheet duty = model	9,598 kW	model P.Qe1 9,598 kW ±0.5 %	✓	
	E-101 clean capacity at full P-101 flow	14,619 kW	≥ 1.2 × 9,598 kW (typical)	✓	
	E-101 fouling margin	UA may fall to 45 % (R_f 1.25 m²K/kW)	typical design fouling 0.35 m²K/kW → 74 %	✓	
	E-101 at the 120 % case	needs 62 % of clean UA	≤ 100 %	✓	
	E-103 rated capacity vs HMB duty	10,086 kW (U 0.45 × 362 m² × F × LMTD)	≥ 1.1 × 7,872 kW	✓	
	E-103 at the 120 % case	10,296 kW	≥ 9,358 kW	✓	
	E-106 rated capacity vs HMB duty	5,085 kW (U 0.4 × 342 m² × F × LMTD)	≥ 1.1 × 3,492 kW	✓	
	E-106 at the 120 % case	4,615 kW	≥ 4,192 kW	✓	
Tube rupture (10/13 rule)	E-101	low side 12 bar g	≥ 10/13 × 40 = 30.8 bar g, else a relief case	✗	EQ-em-3
	E-103	low side 10 bar g	≥ 10/13 × 36 = 27.7 bar g, else a relief case	✗	EQ-em-3

Check	Item	Value	Limit / reference	OK	Query
	E-106	low side 10 bar g	≥ 10/13 × 36 = 27.7 bar g, else a relief case	✗	EQ-em-3
Motor rating ≥ absorbed power	M-101	250 kW	≥ 204 kW (model max / estimate)	✓	
	M-102	not in the load list	CHX-V1-EE-LL-001	✗	EQ-em-7
	P-100A/B	110 kW	≥ 90 kW (model max / estimate)	✓	
	P-101A/B	200 kW	≥ 160 kW (model max / estimate)	✓	
	P-102A/B	37 kW	≥ 30 kW (model max / estimate)	✓	
	K-101	1600 kW	≥ 1,560 kW (model max / estimate)	✓	
Agitator M-101	ungassed power at the datasheet N and D	179 kW at 128 rpm	model K.agitKW 180 kW ±5 %	✓	
	gassed / ungassed ratio (JI-101)	0.85	0.7–0.9 typical for concave-blade gas-dispersing disc turbine	✓	
	not flooded	Fl 0.096	< 2.14 (typical correlation)	✓	
Hot hydrogen (API 941, schematic)	R-101 (liquid in equilibrium with the gas)	200 °C (trip 220) at pH_2 13.8 bar	carbon-steel curve 289 − 28 °C	✓	
	R-101 overhead / E-103 inlet	200 °C (trip 220) at pH_2 13.8 bar	carbon-steel curve 289 − 28 °C	✓	
	R-102	160 °C at pH_2 18.6 bar	carbon-steel curve 278 − 28 °C	✓	
	R-103 bed	159 °C (trip 220) at pH_2 18.6 bar	carbon-steel curve 278 − 28 °C	✓	
	E-106 inlet	159 °C at pH_2 18.6 bar	carbon-steel curve 278 − 28 °C	✓	
	V-104 / K-101 suction	45 °C at pH_2 27.6 bar	carbon-steel curve 266 − 28 °C	✓	
	K-101 discharge / sparger	60 °C at pH_2 30.4 bar	carbon-steel curve 264 − 28 °C	✓	
Documents	EQL items on the EPC P&ID	17 of 17	CHX-V1-PID-001	✓	
	datasheets named by the procurement packages	14 of 14	CHX-V1-PC-PKG-001	✓	

9 · Engineering queries

ID	Query	Documents	Proposed resolution
EQ-em-1	R-101 design temperature below the boiling ceiling The R-101 design temperature 250 °C (SL_ASSUMED.Tdes1, also the R-101 line design temperature) is below the temperature the reactor can reach if the trips fail: the boiling ceiling TbC(PI-101) = 258.3 °C, 272.4 °C at the PSV set. SA-516 Gr 70 keeps S = 138 MPa up to 343 °C, so raising the design temperature costs no thickness.	CHX-V1-ME-EQL-001, Safety layers (SL_ASSUMED.Tdes1, R-101 ladder), CHX-V1-PE-LL-001 (R-101 lines 250 °C)	R-101, E-101 tubes, P-101, the R-101 lines and D-102 designed for 280 °C (≥ the ceiling at the PSV set); Safety layers to keep 250 °C as the critical limit of the ladder or move it.
EQ-em-2	E-101 shell: two pressure sources, one PSV In COOL the E-101 shell is the D-101 water side (PSV-109 8 bar g). In HEAT HY-101A/B isolate it and MP steam (10 bar g, line design 12 bar g) feeds it, so PSV-109 does not protect it. This EQL designs the E-101 shell for 12 bar g / 200 °C and keeps D-101 at 8 bar g. Failure of HY-101A/B to close puts MP steam into D-101.	CHX-V1-ME-EQL-001, CHX-V1-PE-LL-001 8"-S-3002-A1, Safety layers SL-R12, EQ-ep-3, P&ID sheet 1	Relief & vent: fire case on the isolated E-101 shell (own PSV or thermal relief) and the HY-101A/B-fail case on PSV-109 (MP steam through HY-101D); or D-101 also 12 bar g with PSV-109 re-set.
EQ-em-3	Tube rupture is a relief case on E-101, E-103 and E-106 The low-pressure side is below 10/13 of the high-pressure side design: E-101 shell 12 vs tubes 40 bar g (slurry and dissolved H_2 into D-101); E-103 and E-106 CW tubes 10 vs shell 36 bar g (hydrogen into the cooling-water system). TSV-113/TSV-106 are thermal reliefs only.	CHX-V1-ME-EQL-001 (E-101/E-103/E-106 sheets), Safety layers SL_IT, Relief & vent (CHX-V1-PE-RLB-001), HAZOP	Relief & vent to size tube-rupture relief (or raise the low-pressure side to 10/13 of the other: E-101 shell 31 bar g, CW sides 28 bar g); CW return gas detection (HAZOP).
EQ-em-4	Nozzle sizes: line list and EPC P&ID differ Nozzles are sized to the line list CHX-V1-PE-LL-001. The EPC P&ID (its own numbering SIZE-SERVICE-NNNN) draws 26 of these connections at another size: R-101 N1 3"-P-1002-B1 vs P&ID 4"-P-1002; R-101 N4 8"-P-1009-B1 vs P&ID 6"-P-1007; R-101 N7 4"-P-1008-B3 vs P&ID 6"-SL-1014; R-102 N1 4"-P-1008-B3 vs P&ID 6"-SL-1014; R-102 N4 4"-P-1010-B3 vs P&ID 6"-SL-2001; R-102 J1 6"-S-3003-A1 vs P&ID 4"-MS-2020; R-103 N3 8"-PG-2005-B2 vs P&ID 10"-P-2005; E-101 N1 16"-P-1005-B3 vs P&ID 12"-SL-1015; E-101 N2 16"-P-1006-B3 vs P&ID 12"-SL-1012; E-101 N3 8"-BFW-3005-A1 vs P&ID 12"-BW-1022; E-101 N4 16"-S-3006-A1 vs P&ID 14"-LS-1023; E-103 N2 8"-PG-2002-B2 vs P&ID 6"-P-1006; E-106 N1 8"-PG-2005-B2 vs P&ID 10"-P-2005; E-106 N3 8"-CW-3009-A1 vs P&ID 10"-CW-2030; E-106 N4 8"-CW-3010-A1 vs P&ID 10"-CW-2030; D-101 N3 8"-BFW-3005-A1 vs P&ID 12"-BW-1022; D-101 N4 16"-S-3006-A1 vs P&ID 14"-LS-1023; V-103 N1 8"-PG-2002-B2 vs P&ID 6"-P-1006; V-103 N3 8"-P-1009-B1 vs P&ID 6"-P-1007; V-104 N2 8"-H-2007-B2 vs P&ID 10"-G-2007; V-104 N3 4"-P-1013-B1 vs P&ID 6"-P-2011; V-104 N4 1-1/2"-H-2011-B2 vs P&ID 2"-G-2010; V-104 N5 6"-H-2010-B1 vs P&ID 8"-G-2009; P-100 D 3"-P-1002-B1 vs P&ID 4"-P-1002; P-102 S 4"-P-1010-B3 vs P&ID 6"-SL-2001; K-101 S 8"-H-2007-B2 vs P&ID 10"-G-2007.	CHX-V1-PID-001 sheets 1–2, CHX-V1-PE-LL-001, all datasheets	P&ID rev 1 to adopt the line-list numbers and sizes (one numbering in the unit); nozzles stay as the line list.
EQ-em-5	Nozzle connections without a line-list line 11 nozzles connect to lines drawn on the P&ID but missing from the line list: R-101 N8 (4"-SL-1004); R-101 N9 (6"-P-1003); R-101 N10 (6"-FL-1040); R-102 N5 (4"-FL-2015); R-102 N6 (8"-FL-2016); R-102 J3 (6"-CW-2022); R-102 J4 (2"-CD-2023); E-101 N6 (3"-CD-1025); D-101 N5 (1"-BD-1026); D-102 N1 (4"-SL-1004); V-104 N6 (8"-FL-2016).	CHX-V1-PE-LL-001, CHX-V1-PID-001, datasheets	Add these lines (catalyst charge, start-up solvent, relief inlets, jacket return and condensate, E-101 condensate, blowdown, R-102 off-gas upstream of EDV-101) to the line list rev 1.
EQ-em-6	PFD equipment sizes differ from the model spans PFD sizes: R-101 Ø3.6 × 4.5 m (Design tab 3.80 × 4.50); V-104 vertical Ø2 × 5 m ≈ 17.8 m³ against a model level span of 30 m³; D-101 Ø1.4 × 4 m ≈ 6.9 m³ against a water span of 24 t (24.9 m³). This EQL sizes from the model: V-104 2.30 × 9.1 m horizontal, D-101 2.40 × 8.4 m.	CHX-V1-PFD-001 (pfspec real sizes), Design tab, model K.V4span / K.Mspan, CHX-V1-ME-EQL-001	PFD rev 1 drawing sizes from the EQL; or reduce the model spans to the PFD vessels (lead: model change, level dynamics change).
EQ-em-7	M-102 agitator on the P&ID, not in the model or the load list R-102 is a slurry reactor and the P&ID draws M-102, but the model has no R-102 agitator (MOTORS) and the load list has no M-102. Kept on the EQL as HOLD with ≈ 23 kW (1.5 kW/m³ typical, assumed) → 30 kW.	CHX-V1-ME-EQL-001, CHX-V1-EE-LL-001, CHX-V1-PID-001 sheet 2, EQ-ec-1, EQ-px-4	Process to confirm R-102 mixing; if agitated, add M-102 to the load list and the C&E (stop on trip?) and to the model MOTORS (lead).

ID	Query	Documents	Proposed resolution
EQ-em-8	D-102 catalyst charging has no pressure source D-102 is filled at ambient pressure under N_2 (8 bar g header) but discharges into R-101 at 30 bar g through HV-117. The design pressure here is the loop 36 bar g because D-102 sees R-101 when HV-117 is open.	CHX-V1-PID-001 sheet 1, CHX-V1-ME-DS-D-102, SOP (catalyst make-up)	Define charging: at start-up below the N_2 pressure only, or pressurise D-102 from the K-101 discharge (then design 42 bar g), or a slurry pump; add the procedure to the SOP.
EQ-em-9	Model heat-transfer capacities above the datasheet equipment E-103: model UA_3 500 kW/K (capacity 47.9 MW at design CW) vs 163 kW/K for the datasheet area at U 0.45 (assumed). R-102 jacket: model UAj 90 kW/K on 26.4 m² of wetted jacket needs U 3.41 kW/m²K (typical jacket 0.3–0.6).	model.js K.UA3, K.UAj; Flexibility tab (capacities); CHX-V1-ME-DS-E-103, -R-102	Either R-102 gets an internal coil (≈ 180 m² at U 0.5) and E-103 the larger area, or the model UA values are reduced to the datasheets (lead: model change; the Flexibility capacities fall).
EQ-em-10	V-104 design temperature vs its outlet lines V-104 is designed for 190 °C here (loss of E-106 cooling: R-103 outlet 158.6 °C + 30 °C, as EQ-ep-12 asks). The line list keeps the V-104 outlets (8"-H-2007-B2, 4"-P-1013-B1, 1-1/2"-H-2011-B2) at 80 °C and the inlet 8"-PG-2006-B2 at 260 °C.	CHX-V1-ME-DS-V-104, CHX-V1-PE-LL-001	Line list rev 1: V-104 outlet lines to the vessel design temperature, or a TSHH on the E-106 outlet that trips the unit (Safety layers).
EQ-em-11	Relief set pressures to follow the equipment design pressures F-102A/B are designed 40 bar g here (P-102 discharge side, EQ-ep-5) but TSV-102F is set at 36 bar g; the R-102 jacket is 12 bar g but TSV-102J is 10 bar g = the MP steam pressure (EQ-ep-4).	Safety layers SL-R15/SL-R16, CHX-V1-ME-DS-F-102, -R-102, Relief & vent	TSV-102F set 40 bar g and TSV-102J 12 bar g (Safety layers / Relief & vent).

10 · Assumptions

• Allowable stresses, joint efficiencies, plate list, B16.5 ratings and the Nelson-type curves are typical values for teaching (take them from the codes in force).
• R-101 H/D 1.2 and R-103 LHSV 3 h^{-1} (Design tab); R-102 H/D 1.2; D-101 L/D 3.5 with LT-109 taps at 20/70 % of D; V-104 L/D 4 with LT-104 taps at 10/75 % of D; V-103 from the PFD drawing; D-102 and F-102 sizes.
• Exchanger U: E-101 0.85 kW/m²K (Design tab), E-103 0.45, E-106 0.40 kW/m²K; area = larger of 1.2 × design and 1.1 × the 120 % case; tubes 25.4/19.05 mm on 1.25 pitch; bundle correlation.
• Agitator: concave-blade gas-dispersing turbine (Np 3.2) at D/T 0.40 + hydrofoil (Np 0.3) at D/T 0.45; flooding correlation Fl = 30·Fr·(D/T)^3.5 (× 2.5 for concave blades); M-102 1.5 kW/m³.
• Weights: steel × 1.15–1.6 allowances, drives and machine packages are order-of-magnitude values.
• Pump efficiencies 0.60/0.70/0.50; compressor k 1.4, η 0.75; FV-101 1.5 bar and lines 0.5 bar for P-100.
• Catalyst bulk density 900 kg/m³; candle flux 1–3 m³/m²·h; water density fit.